

EDA Summary Report

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1. Missing Values — Top 10 & Treatment Suggestions

#	Column	Missing %	Classification	Suggested Treatment
1	jkhr_win rate (기수-마필 조합 승률)	42.9%	structural	Flag indicator + fill with 0 (or low value)
2	hr_prev_ rating (직전 경주 레이팅)	33.7%	structural	Flag indicator + fill with 0 (or low value)
3	hr_winr ate__z (마필 승률 (경주 내 z 점수))	31.5%	structural	Flag indicator + fill with 0 (or low value)
4	rating__ z (레이팅 (경주 내 z 점수))	26.7%	structural	Flag indicator + fill with 0 (or low value)
5	rating (마필 레이팅)	26.5%	structural	Flag indicator + fill with 0 (or low value)
6	rating__ pr (레이팅 (경주 내 백분위))	26.5%	structural	Flag indicator + fill with 0 (or low value)
7	hr_dist_ 	24.5%	structural	Flag indicator + fill with

#	Column	Missing %	Classification	Suggested Treatment
	winrate (같은 거리 승률)			0 (or low value)
8	hr_style _sd (각질 일관성)	19.1%	structural	Flag indicator + fill with 0 (or low value)
9	style_vs_ race (본인 각질 - 경주 평균)	10.0%	structural	Flag indicator + fill with 0 (or low value)
1 0	hr_style (각질 (주행 스타일))	10.0%	structural	Flag indicator + fill with 0 (or low value)

Key findings: - Structural missing: Caused by system design (e.g., rating not assigned to Grade 6 / 2yo horses) - Random missing: No specific group concentration, impute with group median

2. High Correlation Pairs ($|r| \geq 0.8$)

Total pairs with $|r| \geq 0.8$: **62**

#	Feature 1	Feature 2	r	Drop Candidate	Reason
1	buga1	buga2	1. 00 00	buga2	Redunda nt informati on
2	buga2	buga3	1. 00 00	buga3	Redunda nt informati on
3	buga1	buga3	1. 00 00	buga3	Redunda nt informati

#	Feature 1	Feature 2	r	Drop Candidate	Reason
4	chaksun1	chaksun2	1. 00 00	chaksun2	on Redunda nt informati on
5	chaksun2	chaksun5	0. 99 91	chaksun5	Redunda nt informati on
6	chaksun1	chaksun5	0. 99 91	chaksun5	Redunda nt informati on
7	chaksun4	chaksun5	0. 99 64	chaksun5	Redunda nt informati on
8	chaksun1	chaksun3	0. 99 55	chaksun3	Redunda nt informati on
9	chaksun2	chaksun3	0. 99 55	chaksun3	Redunda nt informati on
10	chaksun2	chaksun4	0. 99 20	chaksun4	Redunda nt informati on
11	chaksun1	chaksun4	0. 99 20	chaksun4	Redunda nt informati on
12	n_run	dusu	0. 99 10	dusu	More missing values
13	hr_starts	hr_style_n	0. 99 07	hr_starts	More missing values
14	chaksun3	chaksun5	0. 99 06	chaksun5	Redunda nt informati

#	Feature 1	Feature 2	r	Drop Candidate	Reason
15	rating	hr_prev_rating	0.9871	hr_prev_rating	More missing values
16	chaksun3	chaksun4	0.9755	chaksun4	Redundant information
17	hr_last_ord	hr_last_finpct	0.9745	hr_last_finpct	Redundant information
18	age__z	age__pr	0.9713	**age__z**	More missing values
19	train_runs_14__z	train_runs_14__pr	0.9680	**train_runs_14__pr**	Derived (z-score/percentile)
20	wg	hr_last_wg	0.9668	hr_last_wg	More missing values

... and 42 more pairs (see *high_correlation_pairs.csv*)

Recommendations: - Consider removing one feature from each highly correlated pair - Prefer keeping the feature with less missing and more interpretability - __z and __pr suffixed columns are standardized versions — keep the original if redundant

3. VIF Analysis (Supplementary — Beyond Textbook Scope)

Note: VIF is NOT covered in the textbook. This is a supplementary reference.

Features with VIF >= 10: **50**

#	Feature	VIF
1	chaksun5	inf
2	buga2	inf
3	chaksun4	inf
4	chaksun1	inf

#	Feature	VIF
5	chaksun3	inf
6	chaksun2	inf
7	buga1	inf
8	buga3	inf
9	wg	1547324.7
10	hr_last_wg	1544405.9
11	dusu	6637.3
12	n_run	6619.7
13	wgBudam	2265.7
14	rating	764.9
15	hr_prev_rating	631.1

Summary of Actions

1. **Missing values:** Apply structural/random classification to decide imputation strategy
2. **High correlation:** Drop one from each pair (preferably the derived or more-missing one)
3. **Before modeling:** Re-check after dropping columns to confirm multicollinearity is resolved